

Lava Client — User Manual

A task-by-task guide to the Lava Android client, derived from the actual feature modules and screens in the repository. UI labels quoted below are taken from the real Compose screens / string resources (source files named per section). Anything not verifiable from the repo is marked UNCONFIRMED:.

Classification: project-specific.

What Lava is

Lava is an unofficial Android client for popular Russian torrent trackers. It ships with a pluggable Tracker SDK that supports multiple content providers (see [Supported providers](#)). You search and browse trackers, view topic detail and comments, download .torrent files or open magnet links, and manage favorites, bookmarks, and history — all from one app. (Source: [README.md](#).)

A companion **Lava API** service does the actual scraping and exposes a JSON API to the app. You can connect the client to a Lava API on your local network, a cloud/remote server, or one running **on this very device** — see [On-device API](#).

Supported providers

Six providers are wired through the Tracker SDK. Verified from each provider's TrackerDescriptor under `core/tracker/<id>/`:

Provider (display name)	Tracker id	Auth type	Notes
RuTracker.org	rutracker	CAPTCHA_LOGIN	Login + captcha (full capability set incl. forum, favorites, upload)
RuTor.info	rutor	FORM_LOGIN	Anonymous by default

Provider (display name)	Tracker id	Auth type	Notes
Internet Archive	archiveorg	NONE	Anonymous — no credentials needed
Project Gutenberg	gutenberg	NONE	Anonymous — no credentials needed
Kinozal.tv	kinozal	FORM_LOGIN	Credentials
NNM-Club	nnmclub	FORM_LOGIN	Credentials

Auth types come from `AuthType (core/tracker/api/.../AuthType.kt)`: `NONE`, `FORM_LOGIN`, `CAPTCHA_LOGIN`, `OAuth`, `API_KEY`. A `NONE` provider is always anonymous; a `FORM_LOGIN/CAPTCHA_LOGIN` provider that supports `Anonymous` shows an anonymous toggle, otherwise it requires credentials (`TrackerDescriptor.kt` KDoc table).

UNCONFIRMED: the `README.md` "Supported trackers" table lists only `RuTracker` + `RuTor` (it dates to 1.2.0). The six descriptors above are present in the source tree today; the `README` table is stale relative to them.

Install / setup

Lava is distributed to testers via **Firebase App Distribution** (the project's distribution channel per the §6.P / §6.Z release process and the distribute logs under `.lava-ci-evidence/distribute-changelog/`). You receive a Firebase invitation, accept it, and install the APK from the link.

UNCONFIRMED: `README.md` also shows Google Play / GitHub / RuStore badges for the client app. The active, repo-verified distribution path is `Firebase App Distribution`; whether the public store listings are live is not determinable from the repository.

The client comes in two variants you can install side by side: a dev/debug build with the `.dev` application-id suffix and a release build. (Source: root `CLAUDE.md` commands section.)

On-device API install (no Play Store)

If you choose to run the API **on this device**, the client links to the separate **Lava API app**. From onboarding's "On this device" section (or **Settings → Run the API on this device**), the button reads **"Install**

Lava API app" when it is not installed and **"Open Lava API app"** when it is (source: [feature/onboarding/.../steps/ApiSelectionStep.kt](#), lines 204-210). When not installed, the client uses a **Firestore download fallback** — **there are no dead Play-Store links** (source: [docs/CONTINUATION.md](#) linking-feature entry, 2026-06-04).

Onboarding

The onboarding flow is a stepped wizard (source: [feature/onboarding/.../OnboardingScreen.kt](#), steps Welcome → ApiSelection → Providers → Configure → Summary).

1. **Welcome** — brand logo, the heading **"Welcome to Lava"**, a "*N* providers available" line, and a **"Get Started"** button. (Source: [WelcomeStep.kt](#).)
2. **Choose your API** ([ApiSelectionStep](#)) — heading **"Choose your API"** with three sections (source: [ApiSelectionStep.kt](#)):
 - **On your network** — auto-discovers Lava APIs over mDNS. Shows "Searching for APIs on your network..." while scanning, then "Found *N* API(s):" or "No APIs discovered on your network." Tap a discovered entry to run a connectivity probe; on success onboarding advances, on failure it shows "Could not reach this API: ..." with a **"Try again"** button. A **"Search again"** button rescans.
 - **On this device** — **"Install Lava API app"** / **"Open Lava API app"** (see above).
 - **Cloud / remote server** — pick a preset or type a server address into the **"Server address (https://host:port)"** field and tap **"Add server"**.
3. **Pick your providers** ([ProvidersStep](#)) — heading **"Pick your providers"**, "Select one or more content providers to configure." Each provider row shows a colored dot, its display name, its auth type (e.g. "Form Login"), and a checkbox. The **"Next"** button is enabled once at least one is selected. (Source: [ProvidersStep.kt](#).)
4. **Configure** — per-provider configuration (credentials / anonymous toggle). See [Provider configuration](#).
5. **Summary** — review and finish.

Provider configuration

Source: [feature/provider_config/.../ProviderConfigScreen.kt](#).

The provider-config screen composes sections including a **Sync** section and — only when `descriptor.supportsAnonymous == true` — an **Anonymous** section toggle. Credentials for login-based providers are entered through the credentials / login screens (see [Login](#)).

Search

Source: [feature/search_input/.../SearchInputScreen.kt](#), [feature/search_result/.../SearchResultScreen.kt](#).

1. Open the search input. It shows a search field (the hint comes from the `designsystem.designsystem_hint_search` string) with a clear (×) action.
2. A **provider chip bar** (`ProviderChipBar`) lets you toggle which configured providers the search runs against. (Source: `SearchInputScreen.kt` lines 92-94.)
3. Submit the query to see results in `SearchResultScreen`. Results carry a **filter** affordance and a category row that starts with **"All"** followed by per-category chips. (Source: `SearchResultScreen.kt`.)
4. If a provider requires login for search, instead of a misleading "Nothing found" the results show a **Login required** empty-state with a **Login** button (SP-3.2 change; source comment in `SearchResultScreen.kt` lines 280-293).

Browse / categories

Source: [feature/category/.../CategoryScreen.kt](#), [feature/forum/.../ForumScreen.kt](#), [feature/search_result/.../categories/CategorySelectionScreen.kt](#).

Browse the tracker's category / forum tree. Category and forum entries render by name. If a forum requires login to browse/search, a **"Login required"** dialog appears with **Login** / cancel actions (source: `CategoryScreen.kt` lines 258-271).

View a topic + comments + download

Source: [feature/topic/.../TopicScreen.kt](#) and [feature/topic/src/main/res/values/strings.xml](#).

The topic screen shows the topic title, poster image, and content, with a share action and a paginated comments list. Verified actions:

- **Magnet** button (`topic_action_magnet = "Magnet"`) — opens a magnet dialog with share / open / cancel actions. Shown only when the topic carries a non-blank magnet link.
- **Torrent** button (`topic_action_torrent = "Torrent"`) — downloads the .torrent file. If you are not authorized, a **"Login required"** dialog appears: *"To download the torrent file you need to be authorized. If you do not have an account you can use a magnet link."* (`topics_login_required_for_download`).
- **Download** requires storage permission; if not granted, a rationale dialog **"Grant access to storage?"** is shown (`permission_write_storage_rationale_title`).
- **Add a comment** — opens an "Add a comment" dialog with a "Write your message..." placeholder and send / cancel actions (`topic_add_comment_title / topic_add_comment_placeholder`). Writing comments requires authorization.
- A favorite toggle is available on the topic.

Login

Source: [feature/login/.../LoginScreen.kt](#), [feature/login/.../ProviderLoginScreen.kt](#).

For providers with `FORM_LOGIN` or `CAPTCHA_LOGIN` auth, the login screens let you enter credentials (and, for RuTracker, solve a captcha). Anonymous-capable providers can be used without credentials. Credentials are stored securely and never logged (per §6.H — see [docs/security/README.md](#)).

UNCONFIRMED: exact on-screen field labels for the login screens were not read line-by-line in this pass; the behavioral contract (form login vs. captcha vs. anonymous) is derived from the descriptors and `AuthType`.

Favorites, bookmarks, history

Sources: [feature/favorites/.../FavoritesScreen.kt](#), [feature/bookmarks/.../BookmarksScreen.kt](#), [feature/visited/](#).

- **Favorites** — topics you have favorited (toggled from the topic screen).

- **Bookmarks** — bookmarked forums/categories.
- **History (visited)** — recently visited items.

Each of these has its own periodic **sync** setting in the menu (see below), with configurable sync periods.

Settings menu

Source: [feature/menu/.../MenuScreen.kt](#)

- `feature/menu/src/main/res/values/strings.xml`. Verified rows include:
- **Settings** section: **Theme** selector; the connection/endpoint selection item; "**Run the API on this device**" row; "**Provider Credentials**"; per-feature sync periods (Favorites / Bookmarks / History / Credentials).
- **Data** section: clear history, clear bookmarks, clear favorites.
- **Misc** section: rights, privacy, contacts, **About**.
- A **Sign out** confirmation flow.

On-device API

The client can connect to a Lava API running on your own phone/tablet. See the dedicated guide: [docs/guides/ON_DEVICE_API_USER_GUIDE.md](#) (start/stop/restart, the access key/pairing, mDNS discovery, the persistent notification, privacy notes, and its own troubleshooting table). Not duplicated here.

TV support

Lava ships a `TvActivity` (leanback) entry point alongside the phone/tablet `MainActivity` (source: `app/src/main/AndroidManifest.xml`, `<activity android:name=".TvActivity">` with `LEANBACK_LAUNCHER`). The manifest declares `android.software.leanback` and `android.hardware.touchscreen` as **not required**, so the app installs on TV-class devices.

Deep links

The manifest registers `rutracker.org` deep links for `viewtopic.php`, `viewforum.php`, and `tracker.php` (both `http` and `https`), so tapping a RuTracker URL can open it directly in Lava (source: `app/src/main/AndroidManifest.xml` MainActivity intent filter).

Troubleshooting

Symptom	What to check
Onboarding can't find an API ("No APIs discovered")	Ensure a Lava API is running on the same Wi-Fi and your router allows multicast; tap "Search again" . Or use Cloud / remote server (enter <code>https://host:port</code>) or On this device . (Source: <code>ApiSelectionStep.kt</code> .)
"Could not reach this API" after selecting one	The connectivity probe failed; tap "Try again" or pick a different API.
Search shows Login required	The selected provider needs credentials for search/download — tap Login and sign in (RuTracker also needs a captcha).
.torrent download blocked / "Login required"	Authorize the provider, or use the Magnet link instead (RuTracker download requires login).
"Grant access to storage?" before download	Grant the storage permission so the .torrent can be written to your download folder.
Provider missing from the search chip bar	Configure/enable it in onboarding or Provider Credentials ; only configured providers appear.
Cloud API returns 401	The cloud server must register your client build's identity. (Source: <code>docs/CONTINUATION.md</code> "cloud-search auth lag" entry — a known operator-side step.)

See also

- Security posture: <docs/security/README.md>

- On-device API user guide: [docs/guides/ON_DEVICE_API_USER_GUIDE.md](#)
- SDK developer guide (adding a provider): [docs/sdk-developer-guide.md](#)
- Architecture: [docs/ARCHITECTURE.md](#)
- Local network discovery: [docs/LOCAL_NETWORK_DISCOVERY.md](#)